Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Regulated Finance & Compliance Systems · Python · React · GCP · Docker · AI-Native Workflows · 9yr Production · Remote

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PROFESSIONAL SUMMARY

Software engineer with 9+ years of production experience building systems that handle complex federal, state, and agency regulations at scale - the exact engineering challenge at the core of Valon's mortgage servicing platform. At Entrata I owned the HUD affordable housing compliance infrastructure for 7 years: annual federal income-limit imports (few thousand source rows → 180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations), LIHTC compliance workflows, and regulatory pipelines for 100K+ housing units with zero compliance violations across full ownership. The engineering discipline required to get regulated financial calculations right - where a wrong import affects thousands of residents' eligibility - is directly transferable to mortgage servicing. LeaseI demonstrates AI-native thinking in a regulated domain: an agentic AI platform that combines RAG over regulatory documents, 8-agent investigation workflows, and deterministic HITL guardrails where LLM reasons but human or deterministic logic makes every compliance call. AgentProbe shows the evaluation infrastructure that makes AI-native systems trustworthy. Python primary stack, React/TypeScript frontend, GCP Cloud Storage, Docker, distributed systems. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages, Frameworks & AI: Python (9yr), TypeScript, JavaScript, React.js, FastAPI, Flask, LangGraph, LangChain, LLM Integration, RAG Pipelines, Agentic Systems, Multi-Agent Orchestration, ChromaDB, Vector Databases, Embeddings, HITL Guardrails, LLM Evaluation, Prompt Engineering, Anthropic API, OpenAI API, MCP

Backend, Data & APIs: REST APIs, API Design, Async Processing, RabbitMQ, Microservices, Distributed Systems, PostgreSQL, MySQL, Oracle, Redis, SQL, NoSQL, Performance Optimization, Query Optimization, Indexing, Idempotency, Reliability Engineering, ETL, Data Pipelines, Compliance Systems, Regulatory Data Modeling

Cloud, DevOps & Observability: Google Cloud Storage (GCP), AWS (S3, EC2, RDS, Glue, Lambda), Docker, CI/CD (GitHub Actions), Automated Testing, Regression Testing, Observability, Structured Logging, Distributed Tracing, Monitoring, Alerting, Incident Response, Root-Cause Analysis, Design Documents, Technical Mentoring, Code Reviews, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - AI-Native Product Delivery

- Designed and shipped a production AI system (LangGraph agentic pipeline, FastAPI, Python) as sole technical owner - built RAG retrieval layer, evaluation pipeline, CI validation gate, HITL guardrails for regulated decisions, and observability infrastructure; delivered 0→1 in weeks.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977) on GCP and AWS; applied Isolation Forest anomaly detection across 3 research pipelines; designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and distributed systems.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Federal, state & agency compliance systems** - owned the HUD affordable housing income-limit import pipeline for 7 years: few thousand federal source rows expanded to ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units; every release included baseline validation, reconciliation, and rollback preparation because one wrong import affects thousands of residents' eligibility; zero compliance violations across full ownership period. Direct analog to mortgage servicing regulatory infrastructure.
- **Distributed backend at production scale** - RabbitMQ async event-driven processing (retry logic, idempotency guarantees) for gross-rent updates across thousands of tenant records; 200+ PostgreSQL/MySQL/Oracle query optimizations (30% performance gain); Docker microservices redesign (50% scalability improvement); AWS S3 data migration with Libsodium-encrypted pipelines (100% PHI compliance, zero security incidents); GitHub Actions CI/CD; GCP Cloud Storage for research pipeline integration.
- **Full-stack delivery & observability** - shipped 20+ production modules (Python, JavaScript, React.js, PostgreSQL) across 120+ Agile sprints; built structured logging, monitoring, and alerting for regulated production systems; led incident response with RCA; built debugging tools reducing triage time 40%; mentored 5+ engineers; 300+ code reviews; Python/Bash automation eliminating 80% of manual workload.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management modules (Python, PostgreSQL); data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

Leasel: AI-Native Platform for Regulated Compliance - Analog to ValonOS Vision

Agentic · RAG · HITL · Regulated Domain · LangGraph · Production

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · SQLite · MCP · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

AI-native compliance platform for a regulated domain - a working prototype of the architecture Valon is building with ValonOS. **Regulated domain AI:** RAG over regulatory documents (100% citation correctness), LLM-as-inference-engine over structured compliance data, HITL guardrails enforcing human oversight on all regulated decisions (LLM reasons and extracts, deterministic logic makes every final compliance call). **Agentic workflow:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Escalation) automating complex investigation workflows. **Measured outcomes:** 93.6% precision, 87.9% F1 across 1,240 test cases - AI-native compliance systems that measurably work.

AgentProbe: AI Evaluation Platform - Trustworthy AI at Scale

LLM Evaluation · RAG · FastAPI · CI Gate · Observability · Open Source

Python · FastAPI · LangGraph · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic API · OpenAI API | github.com/venkatalakshmiparimala/AgentProbe

Production AI evaluation toolkit - the infrastructure that makes AI-native systems trustworthy before they reach production. 15-endpoint FastAPI backend, React.js + TypeScript frontend, 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency), LLM-as-Judge with pass@k multi-trial harness, CI-integrated regression gate (GitHub Actions) blocking quality degradation before deployment. Structured distributed tracing across all agent steps. Used to evaluate Leasel across 1,240+ compliance test cases - zero regressions.

AI on FHIR: AI-Native Application on GCP/AWS - Regulated Healthcare

Full Stack · GCP · AWS Lambda · HIPAA · LLM · React · PostgreSQL

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · GCP · PostgreSQL · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI application for a regulated healthcare environment - LLM-powered NLP translating natural-language queries (EN/ES) into structured FHIR API calls; HIPAA-compliant RBAC with full audit trail; AWS Lambda + GCP backend services; PostgreSQL data layer; Docker containerization; complete observability stack. Demonstrates AI-native development on cloud infrastructure in a regulated environment. ~60% reduction in manual retrieval time.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017